



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis (MAT221)

Riemann Integrability

Cauchy Criterion, Integration of Continuous Functions

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CONDUCTED BY

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Lower Sums

Lower Sum

Let f be bounded on $[a, b]$, and let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$.

The **lower sum** of f with respect to P is

$$L(f, P) = \sum_{i=1}^n m_i(x_i - x_{i-1}).$$

Upper Sums

Upper Sum

Let f be bounded on $[a, b]$, and let

$$P = \{x_0, x_1, \dots, x_n\}$$

be a partition of $[a, b]$.

The **upper sum** of f with respect to P is

$$U(f, P) = \sum_{i=1}^n M_i(x_i - x_{i-1}).$$

Lower Integral

Lower Integral

Let f be bounded on $[a, b]$.

The **lower integral** of f over $[a, b]$ is

$$\int_a^b f = \inf \{L(f, P) : P \text{ is a partition of } [a, b]\}.$$

Upper Integral

Upper Integral

Let f be bounded on $[a, b]$.

The **upper integral** of f over $[a, b]$ is

$$\overline{\int_a^b} f = \sup \{U(f, P) : P \text{ is a partition of } [a, b]\} .$$

Riemann Integrability

Riemann Integrable Function

A bounded function

$$f : [a, b] \rightarrow \mathbb{R}$$

is called **Riemann integrable** on $[a, b]$ if

$$\underline{\int_a^b} f = \overline{\int_a^b} f.$$

The common value is called the **Riemann integral** of f and is denoted by

$$\int_a^b f(x) dx.$$

Problem

Show directly that the constant function

$$f(x) = k$$

is integrable over any closed interval $[a, b]$. What is $\int_a^b f$?



Cauchy Criterion for Riemann Integrability

💡 Cauchy Criterion

A bounded function f on $[a, b]$ is Riemann integrable if and only if for every $\epsilon > 0$, there exists a partition P of $[a, b]$ such that

$$U(f, P) - L(f, P) < \epsilon.$$

❓ Problem

Consider $f(x) = 2x + 1$ over the interval $[1, 3]$. Let P be the partition consisting of the points $\{1, \frac{3}{2}, 2, 3\}$.

- (a) Compute $L(f, P)$, $U(f, P)$, and $U(f, P) - L(f, P)$.
- (b) What happens to the value of $U(f, P) - L(f, P)$ when we add the point $\frac{5}{2}$ to the partition?
- (c) Find a partition P' of $[1, 3]$ for which

$$U(f, P') - L(f, P') < 2.$$

Sequential Criterion

Prove that a bounded function f is integrable on $[a, b]$ if and only if there exists a sequence of partitions $(P_n)_{n=1}^{\infty}$ satisfying

$$\lim_{n \rightarrow \infty} [U(f, P_n) - L(f, P_n)] = 0.$$

🔍 Problem

Using the previous theorem show that the function

$$f(x) = 2x^2 + 3$$

is integrable on $[0, 1]$.



🔍 Problem

Show that the Dirichlet function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q}, \\ 0, & \text{if } x \notin \mathbb{Q}, \end{cases}$$

is not Riemann integrable on $[0, 1]$.

Continuity and Uniform Continuity

Continuity

A function f is continuous on an interval I if for every $c \in I$ and for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $x \in I$, if $|x - c| < \delta$, then $|f(x) - f(c)| < \epsilon$. Here δ may depend on both ϵ and c .

Uniform Continuity

A function f is uniformly continuous on an interval I if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all x, y in the domain of f , if $|x - y| < \delta$, then $|f(x) - f(y)| < \epsilon$.

Continuous Functions are Riemann Integrable

💡 **Continuity $\implies$ Riemann Integrable**

If a function f is continuous on $[a, b]$, then f is Riemann integrable on $[a, b]$.

Problem

Show that if f is increasing on $[a, b]$, then f is Riemann integrable on $[a, b]$.

Thank You!

We'd love your questions and feedback.

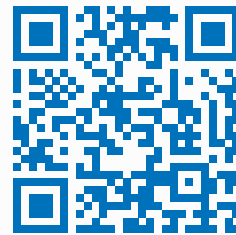
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(Lectures, walkthroughs, and course updates)



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References

- [1] Stephen Abbott, *Understanding Analysis*, 2nd Edition, Springer, 2015.
- [2] Terence Tao, *Analysis I*, 3rd Edition, Texts and Readings in Mathematics, Hindustan Book Agency, 2016.